

KARS, Turkey

WMO: 170980

Lat: 40.564N

Lon: 43.112E

Elev: 5889

StdP: 11.83

Time Zone: 3.00 (E03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-7.8	-16.7	2.8	-12.2	-11.6	3.7	-7.4	27.2	28.5	24.0	27.9	1.4	300	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	29.2	85.8	60.4	82.5	59.9	79.1	59.5	64.7	78.9	62.8	76.6	61.3	75.1	11.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	94.0	72.3	57.4	87.9	70.4	55.8	82.5	68.0	33.2	78.7	31.7	76.6	30.4	74.9	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.2	20.8	17.9	DB	-17.4	90.1	6.7	3.5	-22.3	92.6	-26.2	94.6	-30.0	96.5	-34.9	99.0	(4)
(5)			WB	-17.8	67.7	6.7	3.1	-22.6	69.9	-26.5	71.8	-30.3	73.5	-35.2	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	13.9	17.7	30.4	41.9	50.4	57.7	64.1	65.2	57.1	46.6	33.4	19.9		
	DBStd	18.99	10.18	10.47	9.37	6.31	4.84	4.42	4.45	4.55	5.35	5.82	7.86	10.97		
	HDD50	4508	1120	903	608	250	53	2	0	0	8	134	498	932		
	HDD65	8624	1585	1323	1073	692	452	220	69	53	241	571	948	1397		
	CDD50	1467	0	0	0	8	67	234	437	472	221	28	0	0		
	CDD65	107	0	0	0	0	0	2	41	60	4	0	0	0		
	CDH74	2239	0	0	0	0	3	134	792	1055	253	2	0	0		
	CDH80	571	0	0	0	0	0	10	217	305	39	0	0	0		
	WSAvg	6.2	4.2	4.3	7.4	7.8	7.2	6.9	7.6	7.8	6.5	5.6	5.0	3.9		
	PrecAvg	20.0	0.8	0.9	1.1	2.0	3.1	3.1	2.2	1.7	1.1	1.7	1.0	0.9		
	PrecMax	29.9	2.1	2.4	2.4	4.8	6.3	6.3	5.9	4.0	3.0	4.4	3.2	3.0		
(15) Precipitation	PrecMin	12.7	0.0	0.0	0.1	0.5	1.1	0.7	0.1	0.2	0.0	0.0	0.0	0.1		
	PrecStd	4.6	0.5	0.5	0.6	1.0	1.0	1.4	1.3	1.1	0.8	1.0	0.7	0.6		
	0.4%	DB	37.7	43.3	58.6	68.6	73.0	81.0	89.6	90.0	84.3	72.0	57.7	49.5		
	0.4%	MCWB	33.5	37.9	43.9	48.4	54.0	58.6	62.1	61.1	56.6	51.4	43.2	38.5		
	2%	DB	34.8	37.8	52.2	63.0	69.7	77.6	85.8	86.3	80.3	68.1	53.9	41.2		
(21) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	MCWB	31.8	33.8	40.9	47.0	52.9	58.4	61.3	60.4	55.5	50.4	41.9	34.6		
	5%	DB	32.3	35.5	48.1	59.3	66.3	74.9	82.1	83.7	76.9	64.4	50.2	37.5		
	5%	MCWB	29.9	32.2	39.3	45.8	51.3	58.0	60.9	60.2	55.5	49.4	40.5	32.8		
	10%	DB	30.0	32.4	43.2	55.5	62.9	71.6	78.5	80.4	73.4	60.6	46.5	34.2		
	10%	MCWB	27.8	29.7	37.0	44.3	50.3	56.9	61.0	60.2	54.7	48.0	38.9	30.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.3	38.0	45.8	51.3	57.3	63.7	68.1	68.0	62.0	55.1	45.8	39.0		
	0.4%	MCDB	36.2	42.3	54.6	61.2	69.5	75.5	82.9	83.5	77.1	68.3	53.6	45.4		
	2%	WB	32.2	34.4	42.3	48.7	54.7	61.5	65.0	64.7	59.2	52.3	43.5	35.3		
	2%	MCDB	33.9	37.1	50.4	60.0	65.3	73.6	79.0	79.8	74.5	64.3	50.4	39.5		
	5%	WB	30.0	32.1	39.7	46.9	53.0	59.5	63.2	62.8	57.3	50.3	41.5	33.3		
(31) Mean Daily Temperature Range	5%	MCDB	32.0	34.8	45.9	57.0	63.2	71.5	76.5	77.7	72.3	61.4	48.2	36.6		
	10%	WB	27.5	30.0	37.5	45.0	51.4	57.7	61.7	61.1	55.4	48.6	39.8	31.3		
	10%	MCDB	29.7	32.3	43.1	54.1	60.9	69.1	74.9	76.2	69.9	58.4	45.7	34.0		
	5% DB	MDBR	19.1	20.5	18.9	22.0	22.7	25.9	27.5	29.2	29.6	24.8	22.4	19.4		
	5% DB	MCDBR	15.7	18.2	24.2	29.2	28.5	30.7	32.9	33.3	35.3	32.0	28.5	22.8		
(36) Clear-Sky Solar Irradiance	5% DB	MCWBR	12.8	14.4	14.2	15.4	14.5	14.5	13.5	12.6	15.4	16.5	17.6	16.3		
	5% WB	MCDBR	12.9	16.2	20.5	25.3	24.9	27.9	28.8	29.4	31.8	27.3	24.8	19.1		
	5% WB	MCWBR	11.0	13.1	12.9	14.4	14.0	15.0	14.1	14.2	15.5	14.7	16.0	14.3		
	taub		0.261	0.292	0.341	0.393	0.371	0.349	0.382	0.375	0.323	0.329	0.282	0.259		
	taud		2.312	2.235	2.233	2.158	2.314	2.403	2.330	2.340	2.468	2.405	2.480	2.405		
(42) All-Sky Solar Radiation	Ebn at Noon		287	292	288	279	287	293	282	281	292	276	279	280		
	Edh at Noon		28	35	40	46	40	37	40	38	31	30	23	23		
	RadAvg		802	1174	1481	1571	1791	2183	2229	2034	1648	1079	729	602		
(45)	RadStd		50	80	147	154	124	141	117	80	103	83	92	73		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	-1.69	-1.69	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	N/A	N/A	-0.85	N/A	N/A	N/A	+131	N/A

Nomenclature: See separate page